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Mini Relatório - Sistemas Distribuídos
Prática 2

No wireshark, utilizou-se o filtro ‘ip.addr == 127.0.0.1’
Ao rodar o nameservice e fazer associação com server:

ip.addr == 127.0.0.1

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	127.0.0.1	127.0.0.1	TCP	76	33086 → 9090 [SYN] Seq=0 Win=65495 Len=0 MSS=65495 SACK_PERM TSval=1979377941 TSecr=0 WS=128
2	0.000007709	127.0.0.1	127.0.0.1	TCP	76	9090 → 33086 [SYN, ACK] Seq=0 Ack=1 Win=65483 Len=0 MSS=65495 SACK_PERM TSval=1979377941 TSecr=1979377941 WS=128
3	0.000013467	127.0.0.1	127.0.0.1	TCP	68	33086 → 9090 [ACK] Seq=1 Ack=1 Win=65536 Len=0 TSval=1979377941 TSecr=1979377941
4	0.000122962	127.0.0.1	127.0.0.1	TCP	182	33086 → 9090 [PSH, ACK] Seq=1 Ack=1 Win=65536 Len=114 TSval=1979377941 TSecr=1979377941
5	0.000125694	127.0.0.1	127.0.0.1	TCP	68	9090 → 33086 [ACK] Seq=1 Ack=115 Win=65408 Len=0 TSval=1979377941 TSecr=1979377941
6	0.000405579	127.0.0.1	127.0.0.1	TCP	276	9090 → 33086 [PSH, ACK] Seq=1 Ack=115 Win=65536 Len=208 TSval=1979377941 TSecr=1979377941
7	0.000446859	127.0.0.1	127.0.0.1	TCP	68	33086 → 9090 [ACK] Seq=115 Ack=209 Win=65408 Len=0 TSval=1979377941 TSecr=1979377941
8	0.000683610	127.0.0.1	127.0.0.1	TCP	292	33086 → 9090 [PSH, ACK] Seq=115 Ack=209 Win=65536 Len=224 TSval=1979377942 TSecr=1979377941
9	0.000869126	127.0.0.1	127.0.0.1	TCP	138	9090 → 33086 [PSH, ACK] Seq=209 Ack=339 Win=65536 Len=70 TSval=1979377942 TSecr=1979377942
10	0.050449810	127.0.0.1	127.0.0.1	TCP	68	33086 → 9090 [ACK] Seq=339 Ack=279 Win=65536 Len=0 TSval=1979377992 TSecr=1979377942

Depois de capturar proxy bind:

18	19.467215367	127.0.0.1	127.0.0.1	TCP	76	49972 → 9090 [SYN] Seq=0 Win=65495 Len=0 MSS=65495 SACK_PERM TSval=1979480114 TSecr=0 WS=128
19	19.467222877	127.0.0.1	127.0.0.1	TCP	76	9090 → 49972 [SYN, ACK] Seq=0 Ack=1 Win=65483 Len=0 MSS=65495 SACK_PERM TSval=1979480114 TSecr=1979480114 WS=128
20	19.467228436	127.0.0.1	127.0.0.1	TCP	68	49972 → 9090 [ACK] Seq=1 Ack=1 Win=65536 Len=0 TSval=1979480114 TSecr=1979480114
21	19.467353762	127.0.0.1	127.0.0.1	TCP	182	49972 → 9090 [PSH, ACK] Seq=1 Ack=1 Win=65536 Len=114 TSval=1979480114 TSecr=1979480114
22	19.467356883	127.0.0.1	127.0.0.1	TCP	68	9090 → 49972 [ACK] Seq=1 Ack=115 Win=65408 Len=0 TSval=1979480114 TSecr=1979480114
23	19.467618751	127.0.0.1	127.0.0.1	TCP	276	9090 → 49972 [PSH, ACK] Seq=1 Ack=115 Win=65536 Len=208 TSval=1979480114 TSecr=1979480114
24	19.467640987	127.0.0.1	127.0.0.1	TCP	68	49972 → 9090 [ACK] Seq=115 Ack=209 Win=65408 Len=0 TSval=1979480114 TSecr=1979480114
25	19.467872426	127.0.0.1	127.0.0.1	TCP	184	49972 → 9090 [PSH, ACK] Seq=115 Ack=209 Win=65536 Len=116 TSval=1979480114 TSecr=1979480114
26	19.468063877	127.0.0.1	127.0.0.1	TCP	243	9090 → 49972 [PSH, ACK] Seq=209 Ack=231 Win=65536 Len=175 TSval=1979480115 TSecr=1979480114
27	19.468204125	127.0.0.1	127.0.0.1	TCP	68	49972 → 9090 [FIN, ACK] Seq=231 Ack=384 Win=65536 Len=0 TSval=1979480115 TSecr=1979480115
28	19.468290147	127.0.0.1	127.0.0.1	TCP	68	9090 → 49972 [FIN, ACK] Seq=384 Ack=232 Win=65536 Len=0 TSval=1979480115 TSecr=1979480115
29	19.468294048	127.0.0.1	127.0.0.1	TCP	68	49972 → 9090 [ACK] Seq=232 Ack=385 Win=65536 Len=0 TSval=1979480115 TSecr=1979480115
30	19.468330817	127.0.0.1	127.0.0.1	TCP	76	55720 → 40521 [SYN] Seq=0 Win=65495 Len=0 MSS=65495 SACK_PERM TSval=1979480115 TSecr=0 WS=128
31	19.468334718	127.0.0.1	127.0.0.1	TCP	76	40521 → 55720 [SYN, ACK] Seq=0 Ack=1 Win=65483 Len=0 MSS=65495 SACK_PERM TSval=1979480115 TSecr=1979480115 WS=128
32	19.468338229	127.0.0.1	127.0.0.1	TCP	68	55720 → 40521 [ACK] Seq=1 Ack=1 Win=65536 Len=0 TSval=1979480115 TSecr=1979480115
33	19.468390993	127.0.0.1	127.0.0.1	TCP	203	55720 → 40521 [PSH, ACK] Seq=1 Ack=1 Win=65536 Len=135 TSval=1979480115 TSecr=1979480115
34	19.468393236	127.0.0.1	127.0.0.1	TCP	68	40521 → 55720 [ACK] Seq=1 Ack=136 Win=65408 Len=0 TSval=1979480115 TSecr=1979480115
35	19.468772042	127.0.0.1	127.0.0.1	TCP	281	40521 → 55720 [PSH, ACK] Seq=1 Ack=136 Win=65536 Len=213 TSval=1979480115 TSecr=1979480115
36	19.468796424	127.0.0.1	127.0.0.1	TCP	68	55720 → 40521 [ACK] Seq=136 Ack=214 Win=65408 Len=0 TSval=1979480115 TSecr=1979480115

Quando cliente manda ‘criar’, com os seguintes dados:

```
 /\_/\
( o.o )  What do you want to do?
 > ^ <

[WIRESHARK] Press Enter after capturing packets right after Proxy creatio
n...
[WIRESHARK] Press Enter after capturing packets right after Proxy bind...
c=create, r=read, u=update, d=delete, l=list, q=quit: c
Owner: a
Cat: bb
Days: 12
Price: 1.1
City: cc
Created ID: 4
c=create, r=read, u=update, d=delete, l=list, q=quit:
```

51	80.630636773	127.0.0.1	127.0.0.1	TCP	306	36218 → 39933 [PSH, ACK] Seq=136 Ack=214 Win=65536 Len=238 TSval=1979832316 TSecr=1979785400
52	80.641787884	127.0.0.1	127.0.0.1	TCP	239	39933 → 36218 [PSH, ACK] Seq=214 Ack=374 Win=65536 Len=171 TSval=1979832327 TSecr=1979832316
53	80.641796908	127.0.0.1	127.0.0.1	TCP	68	36218 → 39933 [ACK] Seq=374 Ack=385 Win=65408 Len=0 TSval=1979832327 TSecr=1979832327

E abrindo o pacote de númeo 51:

Wireshark · Packet 51 · any

```
▸ Frame 51: 306 bytes on wire (2448 bits), 306 bytes captured (2448 bits) on interface any, id 0
▸ Linux cooked capture v1
▸ Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
▾ Transmission Control Protocol, Src Port: 36218, Dst Port: 39933, Seq: 136, Ack: 214, Len: 238
    Source Port: 36218
    Destination Port: 39933
    [Stream index: 2]
  ▸ [Conversation completeness: Incomplete, DATA (15)]
    [TCP Segment Len: 238]
    Sequence Number: 136    (relative sequence number)
    Sequence Number (raw): 1926865329
    [Next Sequence Number: 374    (relative sequence number)]
    Acknowledgment Number: 214    (relative ack number)
    Acknowledgment number (raw): 2080680717
    1000 .... = Header Length: 32 bytes (8)
  ▾ Flags: 0x018 (PSH, ACK)
      000. .... .... = Reserved: Not set
      ...0 .... .... = Accurate ECN: Not set
      .... 0... .... = Congestion Window Reduced: Not set
      .... .0.. .... = ECN-Echo: Not set
      .... ..0. .... = Urgent: Not set
      .... ...1 .... = Acknowledgment: Set
      .... .... 1... = Push: Set

0000  00 00 03 04 00 06 00 00  00 00 00 00 00 00 00 08 00   ........ ........
0010  45 00 01 22 42 84 40 00  40 06 f9 4f 7f 00 00 01   E··"B·@· @··O····
0020  7f 00 00 01 8d 7a 9b fd  72 d9 a1 b1 7c 04 ab 0d   ·····z·· r···|···
0030  80 18 02 00 ff 16 00 00  01 01 08 0a 76 01 d7 fc   ........ ····v···
0040  76 01 20 b8 50 59 52 4f  01 f6 04 01 00 00 00 01   v· ·PYRO ........
0050  00 00 00 c6 00 00 00 00  00 00 00 00 00 00 00 00   ........ ........
0060  00 00 00 00 00 00 00 00  00 00 4d c5 23 20 73 65   ........ ··M·# se
0070  72 70 65 6e 74 20 75 74  66 2d 38 20 70 79 74 68   rpent ut f-8 pyth
0080  6f 6e 33 2e 32 0a 28 27  6f 62 6a 5f 66 37 66 33   on3.2·(' obj_f7f3
0090  36 38 36 35 32 36 31 31  34 65 39 62 62 32 35 65   68652611 4e9bb25e
00a0  33 61 35 36 34 35 39 62  30 35 37 32 27 2c 27 63   3a56459b 0572','c
00b0  72 65 61 74 65 5f 62 6f  6f 6b 69 6e 67 27 2c 28   reate_bo oking',(
00c0  7b 27 5f 5f 63 6c 61 73  73 5f 5f 27 3a 27 42 6f   {'__clas s__':'Bo
00d0  6f 6b 69 6e 67 27 2c 27  69 64 27 3a 4e 6f 6e 65   oking',' id':None
00e0  2c 27 6f 77 6e 65 72 5f  6e 61 6d 65 27 3a 27 61   ,'owner_ name':'a
00f0  27 2c 27 63 61 74 5f 6e  61 6d 65 27 3a 27 62 62   ','cat_n ame':'bb
0100  27 2c 27 64 61 79 73 27  3a 31 32 2c 27 70 72 69   ','days' :12,'pri
0110  63 65 5f 70 65 72 5f 64  61 79 27 3a 31 2e 31 2c   ce_per_d ay':1.1,
0120  27 63 69 74 79 27 3a 27  63 63 27 7d 2c 29 2c 7b   'city':' cc'},),{
0130  7d 29                                              })
```

Que possui os mesmos dados enviados pelo cliente

Se fizermos um ‘list’

51	80.036050773	127.0.0.1	127.0.0.1	TCP	360 36218 → 39933 [PSH, ACK] Seq=130 Ack=214 Win=65536 Len=238 TSval=1979832310 TSecr=1979832400
52	80.641787884	127.0.0.1	127.0.0.1	TCP	239 39933 → 36218 [PSH, ACK] Seq=214 Ack=374 Win=65536 Len=171 TSval=1979832327 TSecr=1979832316
53	80.641796908	127.0.0.1	127.0.0.1	TCP	68 36218 → 39933 [ACK] Seq=374 Ack=385 Win=65408 Len=0 TSval=1979832327 TSecr=1979832327
123	336.982490444	127.0.0.1	127.0.0.1	TCP	196 36218 → 39933 [PSH, ACK] Seq=374 Ack=385 Win=65536 Len=128 TSval=1980081855 TSecr=1979832327
124	336.987870900	127.0.0.1	127.0.0.1	TCP	468 39933 → 36218 [PSH, ACK] Seq=385 Ack=502 Win=65536 Len=400 TSval=1980081860 TSecr=1980081855
125	336.987878623	127.0.0.1	127.0.0.1	TCP	68 36218 → 39933 [ACK] Seq=502 Ack=785 Win=65152 Len=0 TSval=1980081860 TSecr=1980081860

Wireshark · Packet 124 · any

```
    [TCP Flags: ·======AP···]
  Window: 512
  [Calculated window size: 65536]
  [Window size scaling factor: 128]
  Checksum: 0xffb8 [unverified]
  [Checksum Status: Unverified]
  Urgent Pointer: 0
  Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
    TCP Option - No-Operation (NOP)
    TCP Option - No-Operation (NOP)
    TCP Option - Timestamps
  [Timestamps]
    [Time since first frame in this TCP stream: 305.098644558 seconds]
    [Time since previous frame in this TCP stream: 0.005380456 seconds]
  [SEQ/ACK analysis]
    [This is an ACK to the segment in frame: 123]
    [The RTT to ACK the segment was: 0.005380456 seconds]
    [iRTT: 0.000007026 seconds]
    [Bytes in flight: 400]
    [Bytes sent since last PSH flag: 400]
  TCP payload (400 bytes)
Data (400 bytes)
  Data [truncated]: 5059524f01f60501004000020000016800000000637dee27638a4e9fa90a192315f236af00004dc5232073657270656e74207574662d38207070974686f6e332e320a5b7b27
```

```
0030  80 18 02 00 ff b8 00 00  01 01 08 0a 76 05 a6 c4   ........ ....v···
0040  76 05 a6 bf 50 59 52 4f  01 f6 05 01 00 40 00 02   v···PYRO ·····@··
0050  00 00 01 68 00 00 00 00  63 7d ee 27 63 8a 4e 9f   ···h···· c}·'c·N·
0060  a9 0a 19 23 15 f2 36 af  00 00 4d c5 23 20 73 65   ···#··6· ··M·# se
0070  72 70 65 6e 74 20 75 74  66 2d 38 20 70 79 74 68   rpent ut f-8 pyth
0080  6f 6e 33 2e 32 0a 5b 7b  27 5f 5f 63 6c 61 73 73   on3.2·[{ '__class
0090  5f 5f 27 3a 27 42 6f 6f  6b 69 6e 67 27 2c 27 69   __':'Boo king','i
00a0  64 27 3a 31 2c 27 6f 77  6e 65 72 5f 6e 61 6d 65   d':1,'ow ner_name
00b0  27 3a 27 72 65 69 27 2c  27 63 61 74 5f 6e 61 6d   ':'rei', 'cat_nam
00c0  65 27 3a 27 6e 65 6e 65  27 2c 27 64 61 79 73 27   e':'nene ','days'
00d0  3a 33 2c 27 70 72 69 63  65 5f 70 65 72 5f 64 61   :3,'pric e_per_da
00e0  79 27 3a 31 30 2e 30 2c  27 63 69 74 79 27 3a 27   y':10.0, 'city':'
00f0  63 77 62 27 7d 2c 7b 27  5f 5f 63 6c 61 73 73 5f   cwb'},{' __class_
0100  5f 27 3a 27 42 6f 6f 6b  69 6e 67 27 2c 27 69 64   _':'Book ing','id
0110  27 3a 33 2c 27 6f 77 6e  65 72 5f 6e 61 6d 65 27   ':3,'own er_name'
0120  3a 27 65 75 27 2c 27 63  61 74 5f 6e 61 6d 65 27   :'eu','c at_name'
0130  3a 27 62 61 6e 61 6e 61  27 2c 27 64 61 79 73 27   :'banana ','days'
0140  3a 31 31 2c 27 70 72 69  63 65 5f 70 65 72 5f 64   :11,'pri ce_per_d
0150  61 79 27 3a 32 31 2e 31  2c 27 63 69 74 79 27 3a   ay':21.1 ,'city':
0160  27 74 6f 6c 65 74 6f 27  7d 2c 7b 27 5f 5f 63 6c   'toleto' },{'__cl
0170  61 73 73 5f 5f 27 3a 27  42 6f 6f 6b 69 6e 67 27   ass__':' Booking'
0180  2c 27 69 64 27 3a 34 2c  27 6f 77 6e 65 72 5f 6e   ,'id':4, 'owner_n
0190  61 6d 65 27 3a 27 61 27  2c 27 63 61 74 5f 6e 61   ame':'a' ,'cat_na
01a0  6d 65 27 3a 27 62 62 27  2c 27 64 61 79 73 27 3a   me':'bb' ,'days':
01b0  31 32 2c 27 70 72 69 63  65 5f 70 65 72 5f 64 61   12,'pric e_per_da
01c0  79 27 3a 31 2e 31 2c 27  63 69 74 79 27 3a 27 63   y':1.1,' city':'c
01d0  63 27 7d 5d                                        c'}]
```

No.: 124 · Time: 336.987870900 · Source: 127.0.0.1 · Destination: 127.0.0.1 · Protocol: TCP · Length: 468 · Info: 39933 → 36218 [PSH, ACK] Seq=385 Ack=502 Win=65536 Len=400 TSval=1980081860 TSecr=1980081855

☑ Show packet bytes

E quando o cliente sai:

124	336.987870900	127.0.0.1	127.0.0.1	TCP	468 39933 → 36218 [PSH, ACK] Seq=385 Ack=502 Win=65536 Len=400 TSval=1980081860 TSecr=1980081855
125	336.987878623	127.0.0.1	127.0.0.1	TCP	68 36218 → 39933 [ACK] Seq=502 Ack=785 Win=65152 Len=0 TSval=1980081860 TSecr=1980081860
147	427.656285178	127.0.0.1	127.0.0.1	TCP	68 36218 → 39933 [FIN, ACK] Seq=502 Ack=785 Win=65536 Len=0 TSval=1980170753 TSecr=1980081860
148	427.656572695	127.0.0.1	127.0.0.1	TCP	68 39933 → 36218 [FIN, ACK] Seq=785 Ack=503 Win=65536 Len=0 TSval=1980170753 TSecr=1980170753
149	427.656578645	127.0.0.1	127.0.0.1	TCP	68 36218 → 39933 [ACK] Seq=503 Ack=786 Win=65536 Len=0 TSval=1980170753 TSecr=1980170753

Por fim, quando o server sai:

125	336.98787862 3	127.0.0.1	127.0.0.1	TCP	68 36218 → 39933 [ACK] Seq=502 Ack=785 Win=65152 Len=0 TSval=1980081860 TSecr=1980081860
147	427.656285178	127.0.0.1	127.0.0.1	TCP	68 36218 → 39933 [FIN, ACK] Seq=502 Ack=785 Win=65536 Len=0 TSval=1980170753 TSecr=1980081860
148	427.656572695	127.0.0.1	127.0.0.1	TCP	68 39933 → 36218 [FIN, ACK] Seq=785 Ack=503 Win=65536 Len=0 TSval=1980170753 TSecr=1980170753
149	427.656578645	127.0.0.1	127.0.0.1	TCP	68 36218 → 39933 [ACK] Seq=503 Ack=786 Win=65536 Len=0 TSval=1980170753 TSecr=1980170753
165	478.895103811	127.0.0.1	127.0.0.1	TCP	68 56118 → 9090 [FIN, ACK] Seq=339 Ack=279 Win=65536 Len=0 TSval=1980220187 TSecr=1979767943
166	478.895338761	127.0.0.1	127.0.0.1	TCP	68 9090 → 56118 [FIN, ACK] Seq=279 Ack=340 Win=65536 Len=0 TSval=1980220187 TSecr=1980220187
167	478.895344705	127.0.0.1	127.0.0.1	TCP	68 56118 → 9090 [ACK] Seq=340 Ack=280 Win=65536 Len=0 TSval=1980220187 TSecr=1980220187